

Kristen Cookies Company Case

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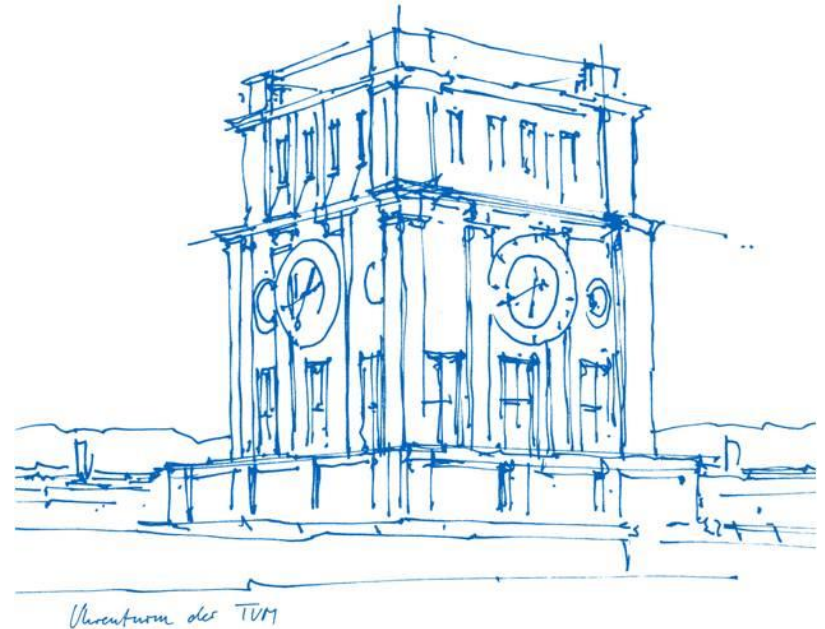
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Understanding Kristen's Cookies Company

Key characteristics of situation and product

- Fresh cookies
- Target hungry students
- Customized cookies
- Make-to-order production
- Risks: Oven failure, order cancellation (not mentioned whether the payment is collected at the order processing stage or the delivery stage. If payment is collected at delivery, the order cancellation is a risk)

Strategy

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A company's strategy is management's action plan for running the business and conducting operations.

It consists of the competitive moves and business approaches that managers are employing to grow the business, attract and please customers, compete successfully, conduct operations, and achieve the targeted levels of organizational performance.

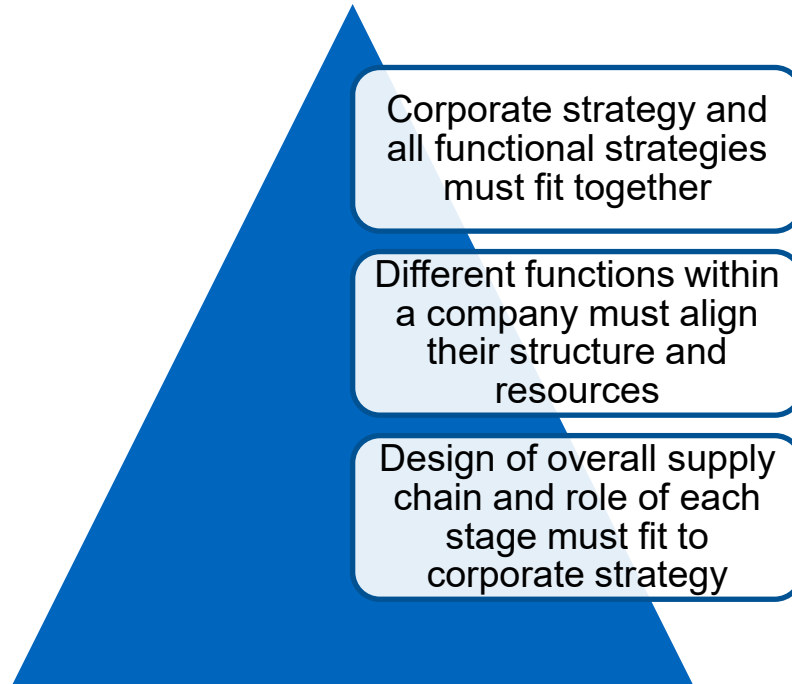
Thompson, Strickland, and Gamble (2007)

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Porter's Generic Strategies

		Strategic Advantage	
		Uniqueness Perceived by the Customer	Low Cost Position
Strategic Target	Industrywide	Differentiation	Overall Cost Leadership
	Particular segment only	Focus	

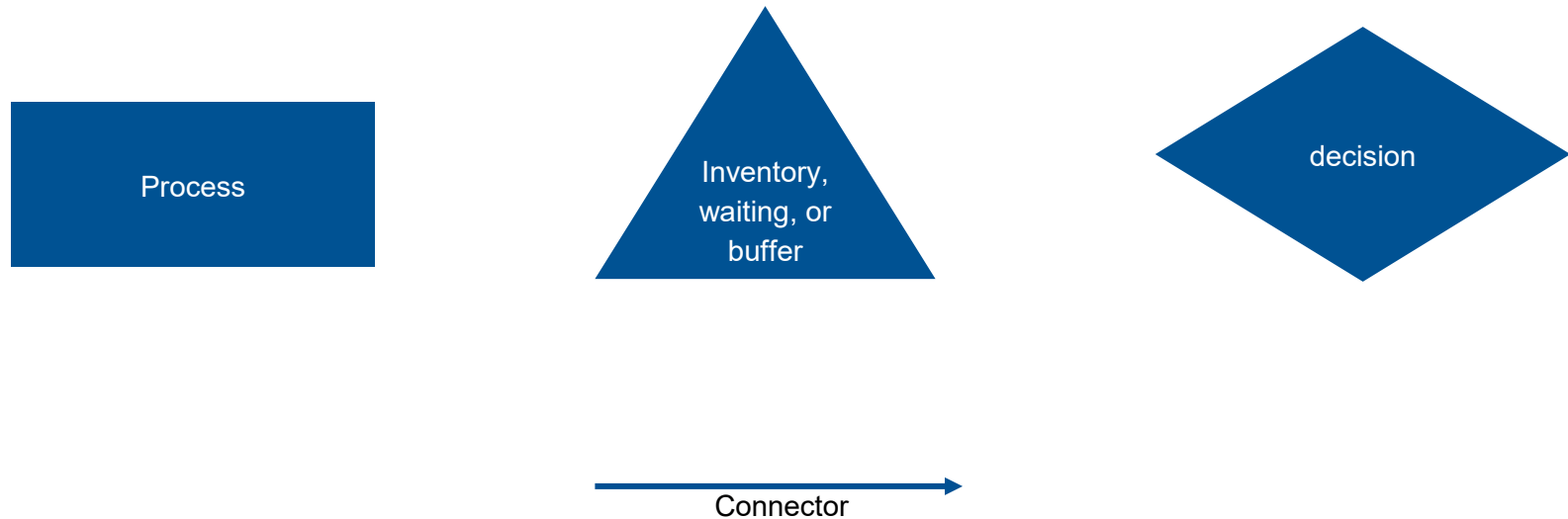
Strategic Fit



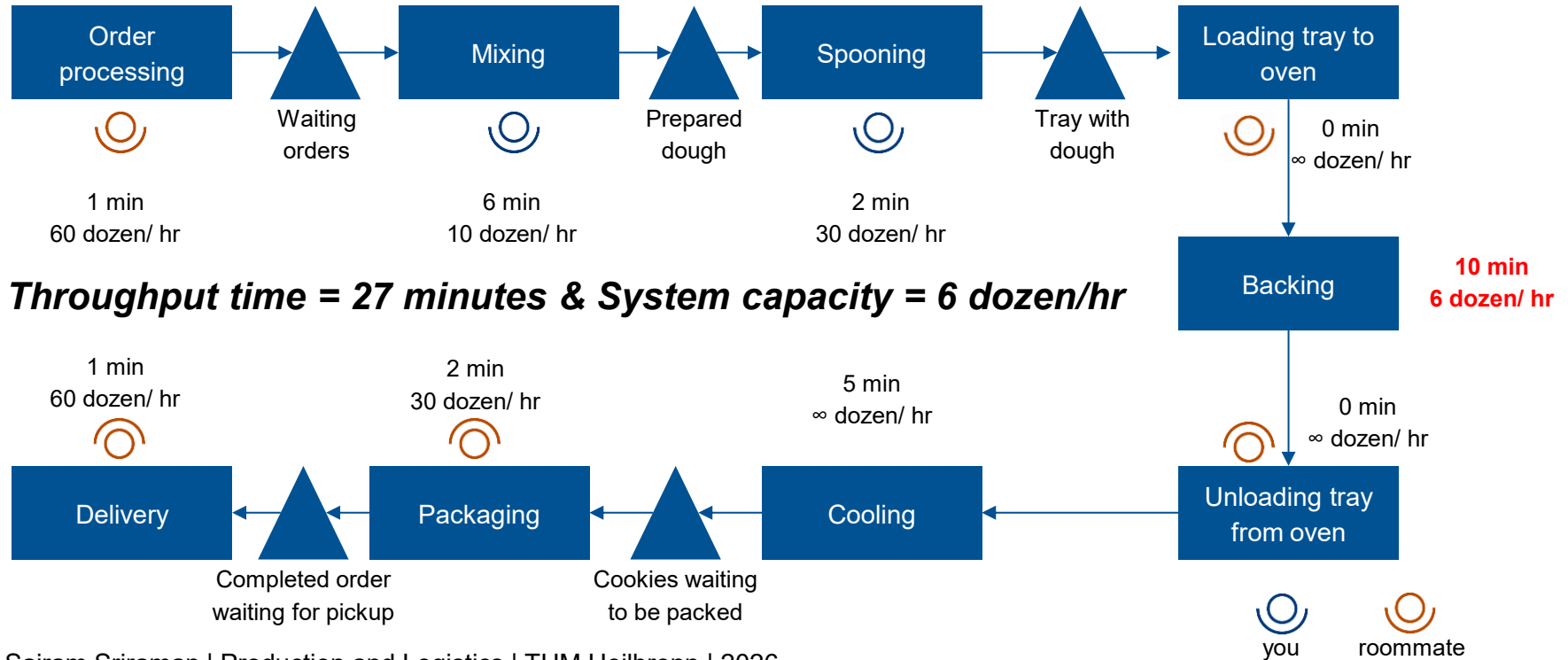
Drivers of issues in the production process

- Different orders cannot be bundled (order pooling is a challenge)
- Limited capacity of the system (oven capacity)
- Limited labor (2 staff and no backup)
- Number of trays on hand
- Order size of the customer per order
- Customer demand
- Availability of raw materials for production (supply constraints)
- In short, flow rate (the number of units that flow through the system in a given time) is the minimum of the customer demand, the system capacity, and the supply inventory. $\text{Flow rate} = \min\{\text{demand rate, capacity, supply rate}\}.$

Elements of Process Flow Diagram

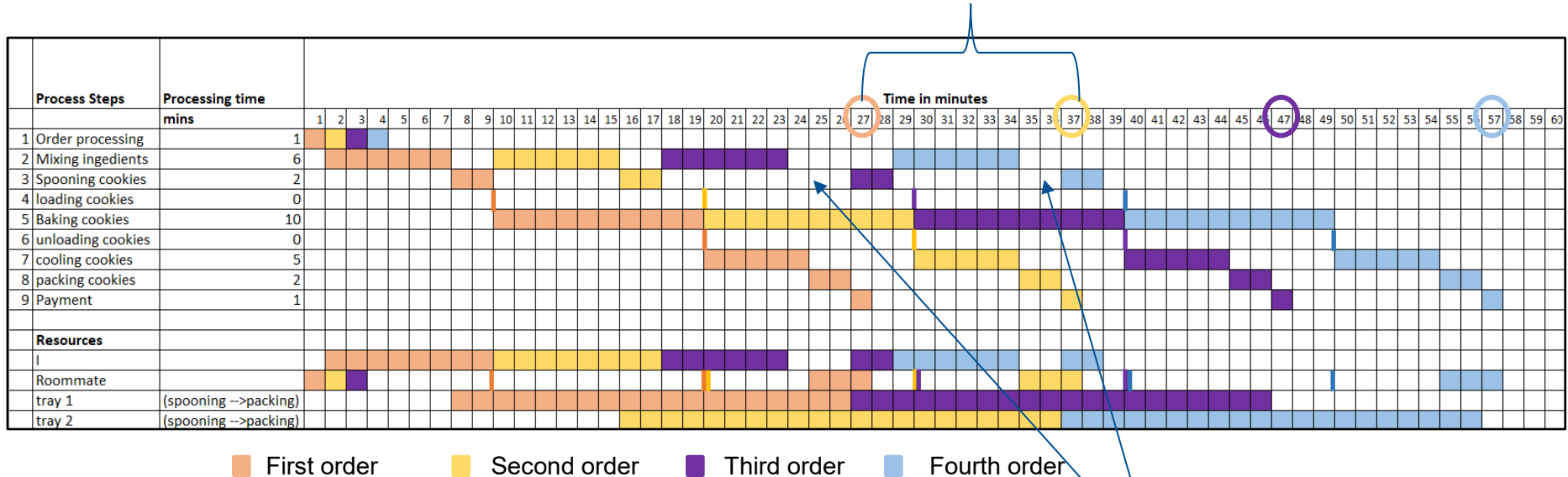


Process Flow Diagram (order size of 1 dozen)



Gantt Chart (1-tray orders, 1 oven)

Cycle time is 10 minutes = Time taken by bottleneck process (Oven)



2 trays are enough for the operation

Dough is waiting for the tray (spooning needs a tray)

Labor costs?

How much is your time worth?

Flexible versus dedicated labor?

Direct labor time per tray

Your time = Mixing time + Spooning time = $6 + 2 = 8$ minutes

Roommate's time = Order processing time + loading time + unloading time + packaging time + delivery time
= $1 + 0 + 0 + 2 + 1 = 4$ minutes

Direct labor time per tray = $8 + 4 = 12$ minutes for 1 tray order size.

For a 2-tray order size (same cookie), mixing, order processing, and delivery take the same time as before, but spooning and packaging take more time. The same goes for a 3-tray order size. However, for a 4-tray order size, you will also have to mix twice (mixer capacity is 3 dozen, which is 3 tray sizes).

Key operations performance indicators

Assume labor cost of \$12 / hour (i.e., \$0.20 / minute). And keep in mind material cost!

Order size (trays)	Throughput time	Cycle time	Direct Labor time	Direct labor time / tray	Direct labor cost / tray	Labor cost / tray (incl. idle)*
1	27	10	12	12	\$2.40	\$4
2	37	10	$12+2+2 = 16$	8	\$1.60	\$4
3	47	10	$12+2\cdot 2+2\cdot 2 = 20$	6.66	\$1.33	\$4
4	57	10	$12+6+3\cdot 2+3\cdot 2 = 30$	7.5	\$1.50	\$4

* Unless you increase flow rate (limited by the cycle time of the bottleneck resource) or figure out a way to get rid of one worker, you'll end up paying 2 workers for 10 (cycle time) at a rate of \$0.20 per minute...

Improvements

What could Kristen Cookies do?

Buy an extra oven. In that case, the baking capacity would increase from 6 dozen/ hr to 12 dozen/hr.

However, the overall system capacity would increase only to 10 dozen/hr, as the new bottleneck will be the mixing process, which takes 6 minutes, assuming flexible labor.

Otherwise, the bottleneck can be “you” (the resource), who takes 8 minutes (mixing + spooning), in which case the system capacity would be 7.5 dozen/ hour. If you are the bottleneck, then the solution is flexible labor.

Also, consider other improvements, such as investing in digital technologies to better forecast customer orders and partnering with food delivery apps like Lieferando, Uber Eats, etc.